NEWTON SCHOOL

MATH · HOMEWORK ANSWER KEY

Estimate, Multiply & Compare

Luna Lee · Grade 4 November 11, 2026

4.NBT.B.5 · Multiply two two-digit numbers using place value and an area model.

Compatible factors are examples. Accept other nearby, easy-to-multiply choices when the estimate, prediction, comparison, and difference agree with the student's factors.

Use the exact products below to check the work. Mixed factor directions do not determine whether an estimate is over or under.

Notes example · 76 × 19

×	10	9
70	70 × 10 = 700	70 × 9 = 630
6	6 × 10 = 60	6 × 9 = 54

Exact: 700 + 630 + 60 + 54 = 1,444.

Method	Estimate	Comparison
Round	80 × 20 = 1,600	Over by 156
Compatible example	75 × 20 = 1,500	Over by 56

Rounding: both factors increase → overestimate.
Compatible: one down, one up → cannot tell from direction alone.

1,600 − 1,444 = 156; 1,500 − 1,444 = 56.
The compatible estimate is closer in this example.

Prediction check

Both positive factors up → over. Both down → under. One unchanged → follow the changed factor. Both unchanged → exact. Mixed directions → cannot tell from direction alone.

1. 28 × 37 = 1,036

Round: 30 × 40 = 1,200; overestimate, difference 164.

Prediction: Both factors increase: overestimate.

Compatible example: 25 × 40 = 1,000; underestimate, difference 36.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	30	7
20	20 × 30 = 600	20 × 7 = 140
8	8 × 30 = 240	8 × 7 = 56

Exact: 600 + 140 + 240 + 56 = 1,036.

Differences: 1,200 − 1,036 = 164; 1,036 − 1,000 = 36.

Compatible estimate is closer.

2. 42 × 63 = 2,646

Round: 40 × 60 = 2,400; underestimate, difference 246.

Prediction: Both factors decrease: underestimate.

Compatible example: 40 × 60 = 2,400; underestimate, difference 246.

Prediction: Both factors decrease: underestimate.

×	60	3
40	40 × 60 = 2,400	40 × 3 = 120
2	2 × 60 = 120	2 × 3 = 6

Exact: 2,400 + 120 + 120 + 6 = 2,646.

Differences: 2,646 − 2,400 = 246; 2,646 − 2,400 = 246.

Both estimates are equally close.

3. 47 × 58 = 2,726

Round: 50 × 60 = 3,000; overestimate, difference 274.

Prediction: Both factors increase: overestimate.

Compatible example: 50 × 60 = 3,000; overestimate, difference 274.

Prediction: Both factors increase: overestimate.

×	50	8
40	40 × 50 = 2,000	40 × 8 = 320
7	7 × 50 = 350	7 × 8 = 56

Exact: 2,000 + 320 + 350 + 56 = 2,726.

Differences: 3,000 − 2,726 = 274; 3,000 − 2,726 = 274.

Both estimates are equally close.

4. 31 × 74 = 2,294

Round: 30 × 70 = 2,100; underestimate, difference 194.

Prediction: Both factors decrease: underestimate.

Compatible example: 30 × 75 = 2,250; underestimate, difference 44.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	70	4
30	30 × 70 = 2,100	30 × 4 = 120
1	1 × 70 = 70	1 × 4 = 4

Exact: 2,100 + 120 + 70 + 4 = 2,294.

Differences: 2,294 − 2,100 = 194; 2,294 − 2,250 = 44.

Compatible estimate is closer.

5. 59 × 22 = 1,298

Round: 60 × 20 = 1,200; underestimate, difference 98.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

Compatible example: 60 × 20 = 1,200; underestimate, difference 98.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	20	2
50	50 × 20 = 1,000	50 × 2 = 100
9	9 × 20 = 180	9 × 2 = 18

Exact: 1,000 + 100 + 180 + 18 = 1,298.

Differences: 1,298 − 1,200 = 98; 1,298 − 1,200 = 98.

Both estimates are equally close.

6. 53 × 39 = 2,067

Round: 50 × 40 = 2,000; underestimate, difference 67.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

Compatible example: 50 × 40 = 2,000; underestimate, difference 67.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	30	9
50	50 × 30 = 1,500	50 × 9 = 450
3	3 × 30 = 90	3 × 9 = 27

Exact: 1,500 + 450 + 90 + 27 = 2,067.

Differences: 2,067 − 2,000 = 67; 2,067 − 2,000 = 67.

Both estimates are equally close.

7. 68 × 41 = 2,788

Round: 70 × 40 = 2,800; overestimate, difference 12.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

Compatible example: 70 × 40 = 2,800; overestimate, difference 12.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	40	1
60	60 × 40 = 2,400	60 × 1 = 60
8	8 × 40 = 320	8 × 1 = 8

Exact: 2,400 + 60 + 320 + 8 = 2,788.

Differences: 2,800 − 2,788 = 12; 2,800 − 2,788 = 12.

Both estimates are equally close.

8. 34 × 27 = 918

Round: 30 × 30 = 900; underestimate, difference 18.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

Compatible example: 35 × 25 = 875; underestimate, difference 43.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	20	7
30	30 × 20 = 600	30 × 7 = 210
4	4 × 20 = 80	4 × 7 = 28

Exact: 600 + 210 + 80 + 28 = 918.

Differences: 918 − 900 = 18; 918 − 875 = 43.

Nearest-ten estimate is closer.

9. 62 × 48 = 2,976

Round: 60 × 50 = 3,000; overestimate, difference 24.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

Compatible example: 60 × 50 = 3,000; overestimate, difference 24.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	40	8
60	60 × 40 = 2,400	60 × 8 = 480
2	2 × 40 = 80	2 × 8 = 16

Exact: 2,400 + 480 + 80 + 16 = 2,976.

Differences: 3,000 − 2,976 = 24; 3,000 − 2,976 = 24.

Both estimates are equally close.

10. 73 × 26 = 1,898

Round: 70 × 30 = 2,100; overestimate, difference 202.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

Compatible example: 75 × 25 = 1,875; underestimate, difference 23.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	20	6
70	70 × 20 = 1,400	70 × 6 = 420
3	3 × 20 = 60	3 × 6 = 18

Exact: 1,400 + 420 + 60 + 18 = 1,898.

Differences: 2,100 − 1,898 = 202; 1,898 − 1,875 = 23.

Compatible estimate is closer.

11. 24 × 38 = 912

Round: 20 × 40 = 800; underestimate, difference 112.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

Compatible example: 25 × 40 = 1,000; overestimate, difference 88.

Prediction: Both factors increase: overestimate.

×	30	8
20	20 × 30 = 600	20 × 8 = 160
4	4 × 30 = 120	4 × 8 = 32

Exact: 600 + 160 + 120 + 32 = 912.

Differences: 912 − 800 = 112; 1,000 − 912 = 88.

Compatible estimate is closer.

12. 49 × 26 = 1,274

Round: 50 × 30 = 1,500; overestimate, difference 226.

Prediction: Both factors increase: overestimate.

Compatible example: 50 × 25 = 1,250; underestimate, difference 24.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	20	6
40	40 × 20 = 800	40 × 6 = 240
9	9 × 20 = 180	9 × 6 = 54

Exact: 800 + 240 + 180 + 54 = 1,274.

Differences: 1,500 − 1,274 = 226; 1,274 − 1,250 = 24.

Compatible estimate is closer.

13. 74 × 19 = 1,406

Round: 70 × 20 = 1,400; underestimate, difference 6.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

Compatible example: 75 × 20 = 1,500; overestimate, difference 94.

Prediction: Both factors increase: overestimate.

×	10	9
70	70 × 10 = 700	70 × 9 = 630
4	4 × 10 = 40	4 × 9 = 36

Exact: 700 + 630 + 40 + 36 = 1,406.

Differences: 1,406 − 1,400 = 6; 1,500 − 1,406 = 94.

Nearest-ten estimate is closer.

14. 51 × 24 = 1,224

Round: 50 × 20 = 1,000; underestimate, difference 224.

Prediction: Both factors decrease: underestimate.

Compatible example: 50 × 25 = 1,250; overestimate, difference 26.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	20	4
50	50 × 20 = 1,000	50 × 4 = 200
1	1 × 20 = 20	1 × 4 = 4

Exact: 1,000 + 200 + 20 + 4 = 1,224.

Differences: 1,224 − 1,000 = 224; 1,250 − 1,224 = 26.

Compatible estimate is closer.

15. 38 × 23 = 874

Round: 40 × 20 = 800; underestimate, difference 74.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

Compatible example: 40 × 25 = 1,000; overestimate, difference 126.

Prediction: Both factors increase: overestimate.

×	20	3
30	30 × 20 = 600	30 × 3 = 90
8	8 × 20 = 160	8 × 3 = 24

Exact: 600 + 90 + 160 + 24 = 874.

Differences: 874 − 800 = 74; 1,000 − 874 = 126.

Nearest-ten estimate is closer.

16. 87 × 26 = 2,262

Round: 90 × 30 = 2,700; overestimate, difference 438.

Prediction: Both factors increase: overestimate.

Compatible example: 90 × 25 = 2,250; underestimate, difference 12.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	20	6
80	80 × 20 = 1,600	80 × 6 = 480
7	7 × 20 = 140	7 × 6 = 42

Exact: 1,600 + 480 + 140 + 42 = 2,262.

Differences: 2,700 − 2,262 = 438; 2,262 − 2,250 = 12.

Compatible estimate is closer.

17. 52 × 48 = 2,496

Round: 50 × 50 = 2,500; overestimate, difference 4.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

Compatible example: 50 × 50 = 2,500; overestimate, difference 4.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	40	8
50	50 × 40 = 2,000	50 × 8 = 400
2	2 × 40 = 80	2 × 8 = 16

Exact: 2,000 + 400 + 80 + 16 = 2,496.

Differences: 2,500 − 2,496 = 4; 2,500 − 2,496 = 4.

Both estimates are equally close.

18. 63 × 24 = 1,512

Round: 60 × 20 = 1,200; underestimate, difference 312.

Prediction: Both factors decrease: underestimate.

Compatible example: 60 × 25 = 1,500; underestimate, difference 12.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	20	4
60	60 × 20 = 1,200	60 × 4 = 240
3	3 × 20 = 60	3 × 4 = 12

Exact: 1,200 + 240 + 60 + 12 = 1,512.

Differences: 1,512 − 1,200 = 312; 1,512 − 1,500 = 12.

Compatible estimate is closer.

19. 95 × 32 = 3,040

Round: 100 × 30 = 3,000; underestimate, difference 40.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

Compatible example: 100 × 30 = 3,000; underestimate, difference 40.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	30	2
90	90 × 30 = 2,700	90 × 2 = 180
5	5 × 30 = 150	5 × 2 = 10

Exact: 2,700 + 180 + 150 + 10 = 3,040.

Differences: 3,040 − 3,000 = 40; 3,040 − 3,000 = 40.

Both estimates are equally close.

20. 67 × 45 = 3,015

Round: 70 × 50 = 3,500; overestimate, difference 485.

Prediction: Both factors increase: overestimate.

Compatible example: 70 × 40 = 2,800; underestimate, difference 215.

Prediction: One factor increases and one decreases: cannot tell from direction alone.

×	40	5
60	60 × 40 = 2,400	60 × 5 = 300
7	7 × 40 = 280	7 × 5 = 35

Exact: 2,400 + 300 + 280 + 35 = 3,015.

Differences: 3,500 − 3,015 = 485; 3,015 − 2,800 = 215.

Compatible estimate is closer.